Dr. Sagar H P

Research Engineer | Data Scientist

sagarhp3589@gmail.com || +49 015560-218813 || linkedin.com/in/hpsagar || hprutvisagar.github.io (e-portfolio)

About me

I hold a Ph.D., from the **Indian Institute of Technology – Bombay.** My exploratory research is focused on integrating **Triboelectric devices for Industry 4.0** applications, which led to the development of self-powering triboelectric vibration sensors for condition and health monitoring of rotary machinery. I have published **8 peer-reviewed journal articles** and **6 conference papers** across multiple fields.

As a seasoned **Research Engineer** and a **Data Scientist**, I wish to leverage my diverse skills to solve complex multi-physics problems to create an impactful success story.

Expertise

- Research and Development
  - Tribo-electric devices
  - condition monitoring and prognostics
  - Electro-mechanical devices
  - Vibration sensors
- Mechanical Engineering
  - 3D modeling
  - Structural Analysis & CFD
  - Design and optimization
- Data Science
  - AI/ML & statistical models
  - Natural Language Processing
  - Failure prediction, Fault/Anomaly detection

EDUCATION

Post-Doctoral Researcher **2024-present**
universität Freiburg
Project: Frequency tunable triboelectric energy harvester

Doctorate of Philosophy **2016-2022**
*Indian Institute of Technology Bombay (**Grade: 8.59 CGPA**)*
Thesis: Triboelectric effect driven self-powered vibration sensors and wind energy harvesting device for enabling industry 4.0.
TATA fellowship awardee

M.Tech - Nanotechnology **2013-2015**
*National Institute of Technology Karnataka, Surathkal (**Grade: 9.5 CGPA**)*
Thesis: Design and analysis of 1D silicon photonic crystal-based strain and mass sensor.
Collaborative research with IISc Bengaluru

B.E. - Mechanical Engineering **2007-2011**
*Visvesvaraya Technological University (**Grade: 71.8%**)*
(Dr. Ambedkar Institute of Technology)
Thesis: Design and development of automated bike washer unit.

WORK EXPERIENCE (6+ years)

Advanced Analytics Specialist **Dec, 2023 - Apr, 2024**
Swiss Re

- Generating analytical & data science solutions to enhance processes and guide business decisions.
- Designed and developed an end-to-end client ranking solution to automate the auditing process.

Data Scientist **June, 2021-Aug, 2023**
Halliburton

- Managed a comprehensive proof-of-concept initiative with complete ownership of backend model development, high-level system designing, and frontend optimizations. I built a resilient regression-based model which can estimate the cloud-application runtime with >95% confidence.

Senior Structural Analyst **June, 2019 - June, 2021**
Gorgonian Technologies

- Here, I was actively involved in research and development of small-scale wind turbines. My major responsibilities were to perform CFD analysis and check the feasibility of the designer's ideas.

Mechanical engineer **Feb, 2019 - Feb, 2020**
Aumeesh Technologies

- Here, I worked on a KAFO (leg prosthetic) product to optimize its mechanical systems based on GAIT analysis.

Mechanical (CAD) Engineer **Sept, 2011 - Aug, 2013**
HCL technologies

- Provided CAD support for semiconductor domain clients. I specialized in harness routing path optimization, and creating its flat-board drawings for manufacturing.

Soft Skills and Strengths

Creativity | Curiosity | Flexibility | Self Confidence | Ability to Plan and Organize | Autonomy | Adaptability | Eye for Details | Problem Solving | Team Working | Love Learning New Things | Leadership | Good Communication | Managing Information | Diplomacy | Good Listener | Patience

BioData

Date of Birth
3rd July 1989

Gender
Male

Marital status
Married

Current address
Dr. Sagar Hosangadi Prutvi,
Sundgaualle 58
Wohnungnummer: 58-00-04
79110 Freiburg, Germany
Ph: +49 015560-218813

Permanent address
#506, Siri Dew Drops Apartments,
7th main Nandakumar layout,
Ramanjaneya Nagara, Arehalli,
Bengaluru - 560061, India
Ph: +91 8660770698

Nationality
Indian

Languages
Kannada - Native
English - Professional fluent
Hindi - Conversational
Telugu - Conversational

SELECTED PUBLICATIONS

Journal Article
Total 8

Self powering vibration sensor based on a cantilever system with a single electrode mode triboelectric nanogenerator, **Sagar Hosangadi Prutvi**, Mallikarjuna Korrapati, and Dipti Gupta, *Measurement Science and Technology 33 (7), 075115 (2022)*, doi 10.1088/1361-6501/ac5b2b

Triboelectric effect based self-powered compact vibration sensor for predictive maintenance of industrial machineries, Hosangadi Prutvi Sagar, Sunil Meti, Udaya K Bhat, and Dipti Gupta*, *Measurement Science and Technology 32 (9), 095119 (2021)*, doi 10.1088/1361-6501/abe6d2

Transient dynamic distributed strain sensing using photonic crystal waveguides, **Hosangadi Prutvi Sagar**, Vignesh Mahalingam, Debiprosad Roy Mahapatra*, Gopalkrishna Hegde, Sathyanarayana Hanagud, and Mohammad Rizwanur Rahman, *Applied Optics 56 (28), 7877-7885 (2017)*, doi 10.1364/AO.56.007877

Conference & symposiums
Total 6

Graphene Integrated Waveguide for Molecular Sensing, **Sagar H P**, and MR Rahman,, *International Engineering Symposium (IES) - 2015,* ***Kumamoto University, Kumamoto, Japan***

SKILLS

Research & Development	Matlab | Rapid Prototyping | SEM & AFM | Thermal Evaporator | Plasma ashers | Screen printing | wet bench experimentations | Instrumentations | Analog circuits | Arduino
Simulation & Analysis	COMSOL Multiphysics | Mathematica | Ansys | Ansa | LS-Dyna | LT-spice | Hypermesh | LAMMPS
3D modeling	Creo | ProE | Autocad | Solidworks | Catia
Data Science	Python | AWS (Sagemaker and Canvas) | Orange | TensorFlow | Keras | Pandas | Seaborn | Numpy

CERTIFICATES

- NLP - Natural Language Processing with Python (*2022*)
- Python for Time Series Data Analysis (*2022*)
- Python for Data Science and Machine Learning bootcamp (*2021*)

- Generative AI fundamentals (*2023*)